

Solid-State Electronics

Chapter 1 & 2 Homework

2024-11-05



Homework 1



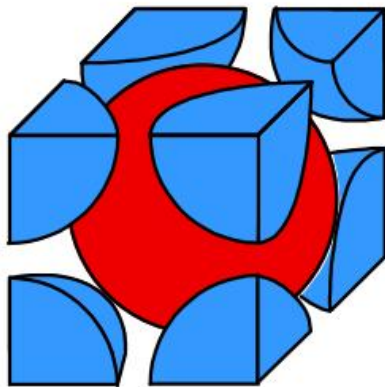
1.22 BCC and FCC crystals

a. Molybdenum has the BCC crystal structure, has a density of 10.22 g cm^{-3} and an atomic mass of 95.94 g mol^{-1} . What is the atomic concentration, lattice parameter a , and atomic radius of molybdenum?

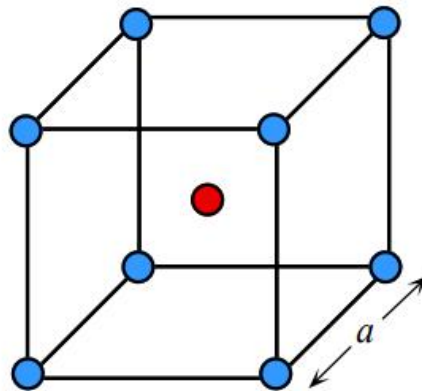
Solution:

$$\rho = \frac{\text{Mass of atoms in unit cell}}{\text{Volume of unit cell}} = \frac{(\text{Number of atoms in unit cell}) \times (\text{Mass of one atom})}{\text{Volume of unit cell}}$$

$$x = 1 + 8 \times \frac{1}{8} = 2$$



a



b

$$\rho = \frac{2 \left(\frac{M_{at}}{N_A} \right)}{a^3}$$



1. lattice parameter a

$$a = \sqrt[3]{\frac{2M_{at}}{\rho N_A}} = \sqrt[3]{\frac{2 \times (95.94 \times 10^{-3} \text{ kg mol}^{-1})}{(10.22 \times 10^3 \text{ kg m}^{-3}) \times (6.022 \times 10^{23} \text{ mol}^{-1})}}$$
$$= 3.147 \times 10^{-10} \text{ m}$$

2. atomic concentration

$$n_{at} = \frac{2}{a^3} = \frac{2}{(3.147 \times 10^{-10} \text{ m})^3} = 6.415 \times 10^{28} \text{ atoms} \cdot \text{m}^{-3}$$

3. atomic radius (P44 Table 1.3)

$$R = \frac{a\sqrt{3}}{4} = \frac{(3.147 \times 10^{-10} \text{ m}) \times \sqrt{3}}{4} = 1.363 \times 10^{-10} \text{ m}$$



1. atomic concentration

$$n_{at} = \frac{\rho \cdot N_A}{M_{at}} = \frac{10.22 \text{ g cm}^{-3} \times 6.022 \times 10^{23} \text{ mol}^{-1}}{95.94 \text{ g mol}^{-1}} \\ = 6.415 \times 10^{28} \text{ atoms} \cdot \text{m}^{-3}$$

2. lattice parameter a

$$a = \sqrt[3]{\frac{2M_{at}}{\rho N_A}} = \sqrt[3]{\frac{2 \times (95.94 \times 10^{-3} \text{ kg mol}^{-1})}{(10.22 \times 10^3 \text{ kg m}^{-3}) \times (6.022 \times 10^{23} \text{ mol}^{-1})}} \\ = 3.147 \times 10^{-10} \text{ m}$$

3. atomic radius (P44 Table 1.3)

$$R = \frac{a \sqrt{3}}{4} = \frac{(3.147 \times 10^{-10} \text{ m}) \times \sqrt{3}}{4} = 1.363 \times 10^{-10} \text{ m}$$



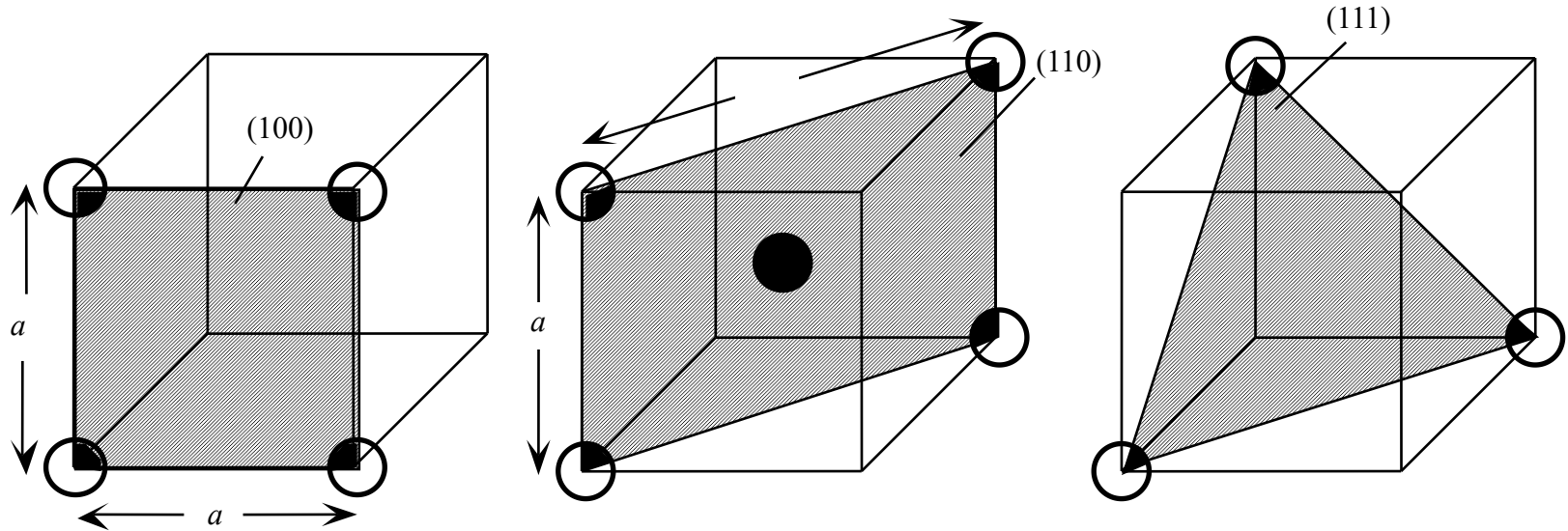
1.24 Planar and surface concentrations

Niobium (Nb) has the BCC crystal with a lattice parameter $a = 0.3294$ nm.

Find the planar concentrations as the number of atoms per nm² of the (100), (110) and (111) planes. Which plane has the most concentration of atoms per unit area? Sometimes the number of atoms per unit area n_{surface} on the surface of a crystal is estimated by using the relation $n_{\text{surface}} = n_{\text{bulk}}^{2/3}$ where n_{bulk} is the concentration of atoms in the bulk. Compare n_{surface} values with the planar concentrations that you calculated and comment on the difference. [Note: The BCC (111) plane does not cut through the center atom and the (111) has one-sixth of an atom at each corner.]

$$n_{(hkl)} = \frac{\text{Number of atoms in unit planar}}{\text{Area of unit planar}} = \frac{x}{S}$$



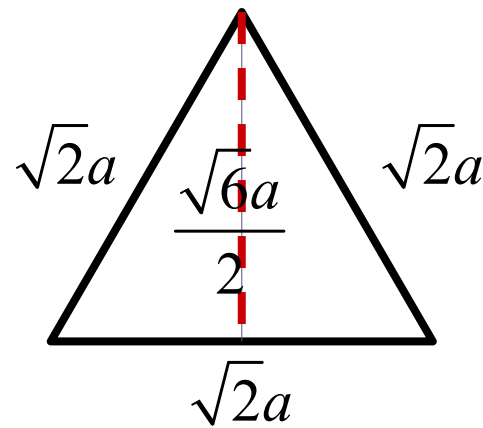
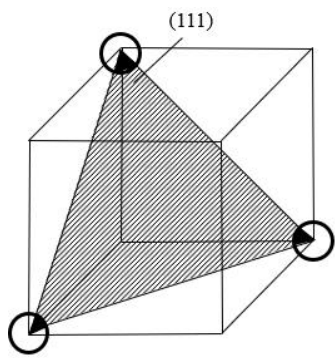


Solution:

$$n_{(100)} = \frac{4 \times \left(\frac{1}{4}\right)}{a^2} = \frac{1}{\left(3.294 \times 10^{-10} \text{ m}\right)^2} = 9.22 \times 10^{18} \text{ atoms} \cdot \text{m}^{-2}$$

$$n_{(110)} = \frac{4 \times \left(\frac{1}{4}\right) + 1}{\sqrt{2}a^2} = \frac{2}{\sqrt{2} \times \left(3.294 \times 10^{-10} \text{ m}\right)^2} = 1.303 \times 10^{19} \text{ atoms} \cdot \text{m}^{-2}$$





$$S = \frac{1}{2} \times \sqrt{2}a \times \frac{\sqrt{6}a}{2} = \frac{\sqrt{3}a^2}{2}$$

$$n_{(111)} = \frac{3 \times \frac{1}{6}}{\frac{a^2 \sqrt{3}}{2}} = \frac{1}{\sqrt{3}a^2} = \frac{1}{\sqrt{3} \times (3.294 \times 10^{-10} \text{ m})^2} = 5.321 \times 10^{18} \text{ atoms} \cdot \text{m}^{-2}$$

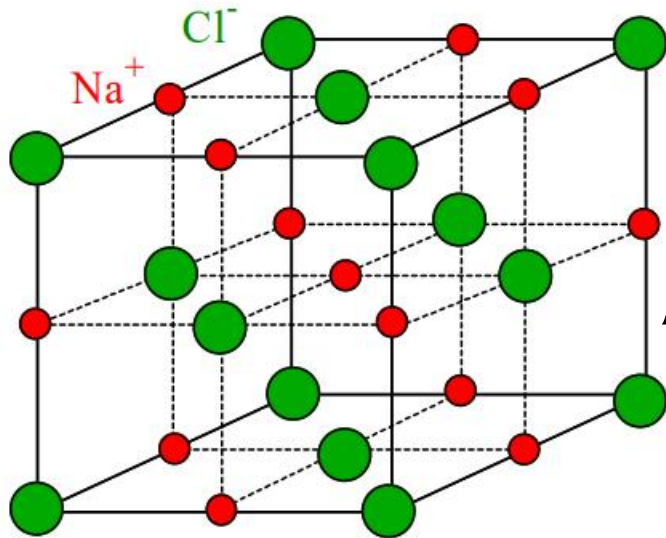
$$n_{bulk} = \frac{2}{a^3} = \frac{2}{(3.294 \times 10^{-10} \text{ m})^3} = 5.596 \times 10^{28} \text{ atoms} \cdot \text{m}^{-3}$$

$$n_{surface} = (n_{bulk})^{\frac{2}{3}} = (5.596 \times 10^{28} \text{ m}^{-3})^{\frac{2}{3}} = 1.463 \times 10^{19} \text{ atoms} \cdot \text{m}^{-2}$$

- $n_{surface}$ is similar to the concentration of the planar surface (110)

1.26 Zinc blende, NaCl and CsCl

b. CdO has the NaCl crystal structure with a lattice parameter of 0.4695 nm. Given the atomic masses of Cd ($112.41 \text{ g mol}^{-1}$) and O (16.00 g mol^{-1}), find the density.



Solution:

$$\begin{aligned}\rho_{\text{CdO}} &= \frac{x \left(\frac{M_{\text{atCd}}}{N_A} \right) + y \left(\frac{M_{\text{atO}}}{N_A} \right)}{a^3} = \frac{x (M_{\text{atCd}} + M_{\text{atO}})}{N_A a^3} \\ &= \frac{4 \times (112.41 \times 10^{-3} \text{ kg mol}^{-1} + 16.00 \times 10^{-3} \text{ kg mol}^{-1})}{(6.022 \times 10^{23} \text{ mol}^{-1}) \times (0.4695 \times 10^{-9} \text{ m})^3} \\ &= 8.24 \times 10^3 \text{ kg} \cdot \text{m}^{-3} = 8.24 \text{ g} \cdot \text{cm}^{-3}\end{aligned}$$



Homework 2



1.31 Pb-Sn solder

Consider the soldering of two copper components. When the solder melts, it wets both metal surfaces. If the surfaces are not clean or have an oxide layer, the molten solder cannot wet the surfaces and the soldering fails. Assume that soldering takes place at **250 °C, and consider the diffusion of Sn atoms into the copper (the Sn atom is smaller than the Pb atom and hence diffuses more easily).**



$$D = D_0 \exp\left(-\frac{qE_A}{kT}\right) \quad \text{P39 Eq. 1.34} \quad L_{rms} = \sqrt{2Dt} \quad \text{P39 Eq. 1.33}$$

- a. The diffusion coefficient of Sn in Cu at two temperatures is $D = 1.69 \times 10^{-9} \text{ cm}^2 \text{ hr}^{-1}$ at 400°C and $D = 2.48 \times 10^{-7} \text{ cm}^2 \text{ hr}^{-1}$ at 650°C . Calculate the **rms distance** diffused by an Sn atom into the copper, assuming the cooling process takes 10 seconds.

Solution:

$$D_1 = 1.69 \times 10^{-9} \text{ cm}^2/\text{hr} = 4.694 \times 10^{-17} \text{ m}^2/\text{s}$$

$$D_2 = 2.48 \times 10^{-7} \text{ cm}^2/\text{hr} = 6.889 \times 10^{-15} \text{ m}^2/\text{s}$$

$$\begin{aligned} D_1 &= D_0 \exp\left(-\frac{qE_A}{kT_1}\right) \\ D_2 &= D_0 \exp\left(-\frac{qE_A}{kT_2}\right) \end{aligned} \quad \frac{D_1}{D_2} = \frac{D_0 \exp\left(-\frac{qE_A}{kT_1}\right)}{D_0 \exp\left(-\frac{qE_A}{kT_2}\right)} = \exp\left(\frac{qE_A}{kT_2} - \frac{qE_A}{kT_1}\right) = \exp\left(\frac{q[T_1 - T_2]E_A}{kT_1T_2}\right)$$



$$E_A = \frac{kT_1T_2 \ln\left(\frac{D_1}{D_2}\right)}{q(T_1 - T_2)}$$

$$= \frac{(1.381 \times 10^{-23} \text{ J K}^{-1}) \times (673 \text{ K}) \times (923 \text{ K}) \times \ln\left(\frac{4.694 \times 10^{-17} \text{ m}^2/\text{s}}{6.889 \times 10^{-15} \text{ m}^2/\text{s}}\right)}{(1.602 \times 10^{-19} \text{ J/eV}) \times (673 \text{ K} - 923 \text{ K})}$$

$$= 1.068 \text{ eV/atom}$$

$$D_o = \frac{D_1}{\exp\left(-\frac{qE_A}{kT_1}\right)} = \frac{4.694 \times 10^{-17} \text{ m}^2/\text{s}}{\exp\left(-\frac{(1.602 \times 10^{-19} \text{ J/eV}) \times (1.068 \text{ eV})}{(1.381 \times 10^{-23} \text{ J K}^{-1}) \times (673 \text{ K})}\right)} = 4.638 \times 10^{-9} \text{ m}^2/\text{s}$$

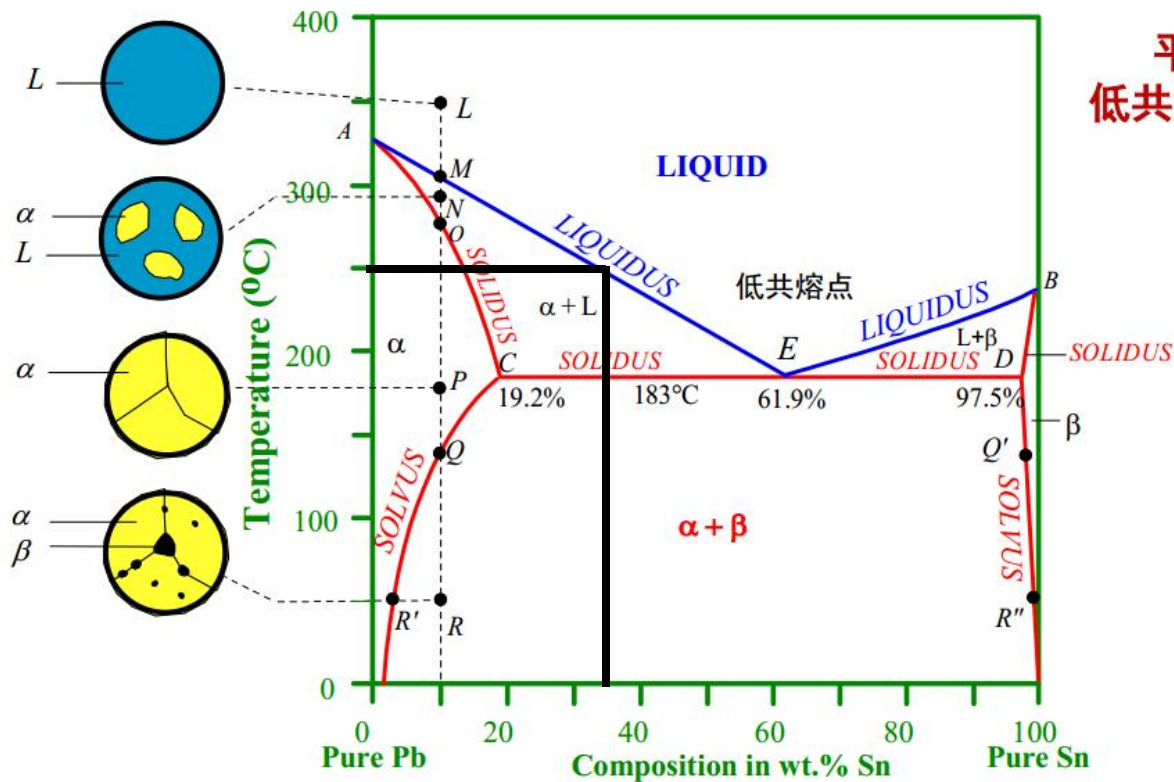
$$D_3 = D_o \exp\left(-\frac{qE_A}{kT_3}\right) = (4.638 \times 10^{-9} \text{ m}^2/\text{s}) \times \exp\left(-\frac{(1.602 \times 10^{-19} \text{ J/eV}) \times (1.068 \text{ eV})}{(1.381 \times 10^{-23} \text{ J K}^{-1}) \times (523 \text{ K})}\right)$$

$$= 2.391 \times 10^{-19} \text{ m}^2/\text{s at } 250^\circ\text{C} (523 \text{ K})$$

$$L_{rms} = \sqrt{2D_3t} = \sqrt{2 \times (2.391 \times 10^{-19} \text{ m}^2/\text{s}) \times (10 \text{ s})} = 2.19 \times 10^{-9} \text{ m}$$



b. What should be the composition of the solder if it is to begin freezing at 250 °C?



平衡
低共熔相图

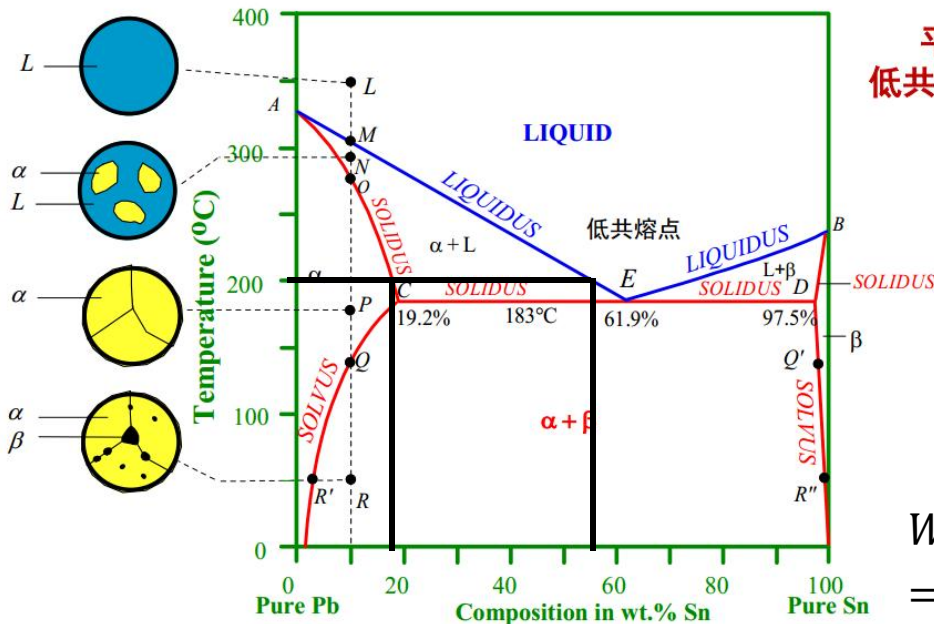
Solution:

$$C_o = 0.33 \text{ (Sn)}$$



c. What are the components (phases) in this alloy at 200 °C? What are the compositions of the phases and their relative weights in the alloy?

Solution: $C_s = 0.18$ and $C_L = 0.56$



平衡
低共熔相图

$$\begin{cases} W_L + W_S = 1 \\ C_L W_L + C_S W_S = C_O \end{cases}$$

P71 Eq.1.36

The weight fraction of α and L phases are:

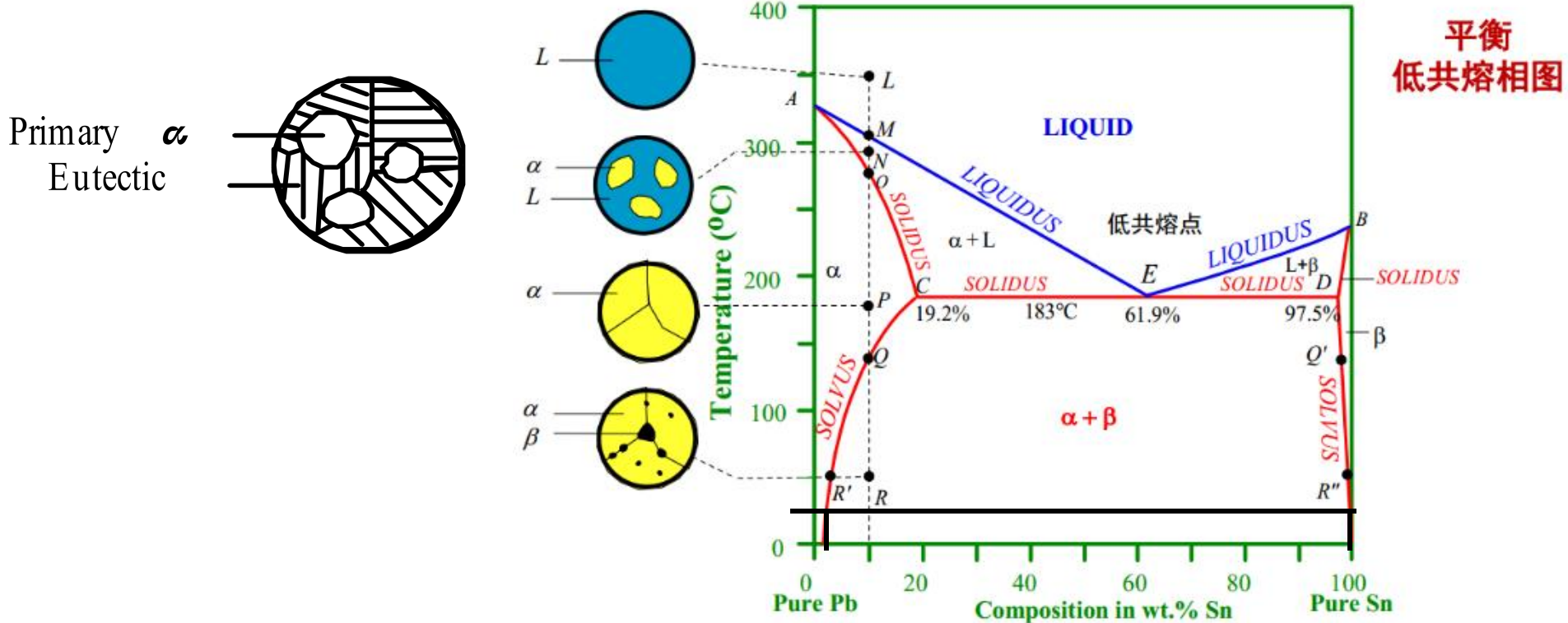
$$W_\alpha = \frac{C_L - C_o}{C_L - C_s} = \frac{0.56 - 0.33}{0.56 - 0.18} = 0.605 \alpha - \text{phase}$$

$$W_L = \frac{C_o - C_s}{C_L - C_s} = \frac{0.33 - 0.18}{0.56 - 0.18} = 0.395 \text{ Liquid-phase}$$



d. What is the microstructure of this alloy at 25 °C? What are weight fractions of the α and β phases assuming near equilibrium cooling?

Solution:



Assuming equilibrium concentrations have been reached:

$$C'_{\alpha} = 0.02 \text{ and } C'_{\beta} = 1$$

The weight fraction of α in the whole alloy is then:

$$W'_{\alpha} = \frac{C'_{\beta} - C_0}{C'_{\beta} - C'_{\alpha}} = \frac{1 - 0.33}{1 - 0.02} = 0.684 \text{ } \alpha \text{-phase}$$

The weight fraction of β in the whole alloy is

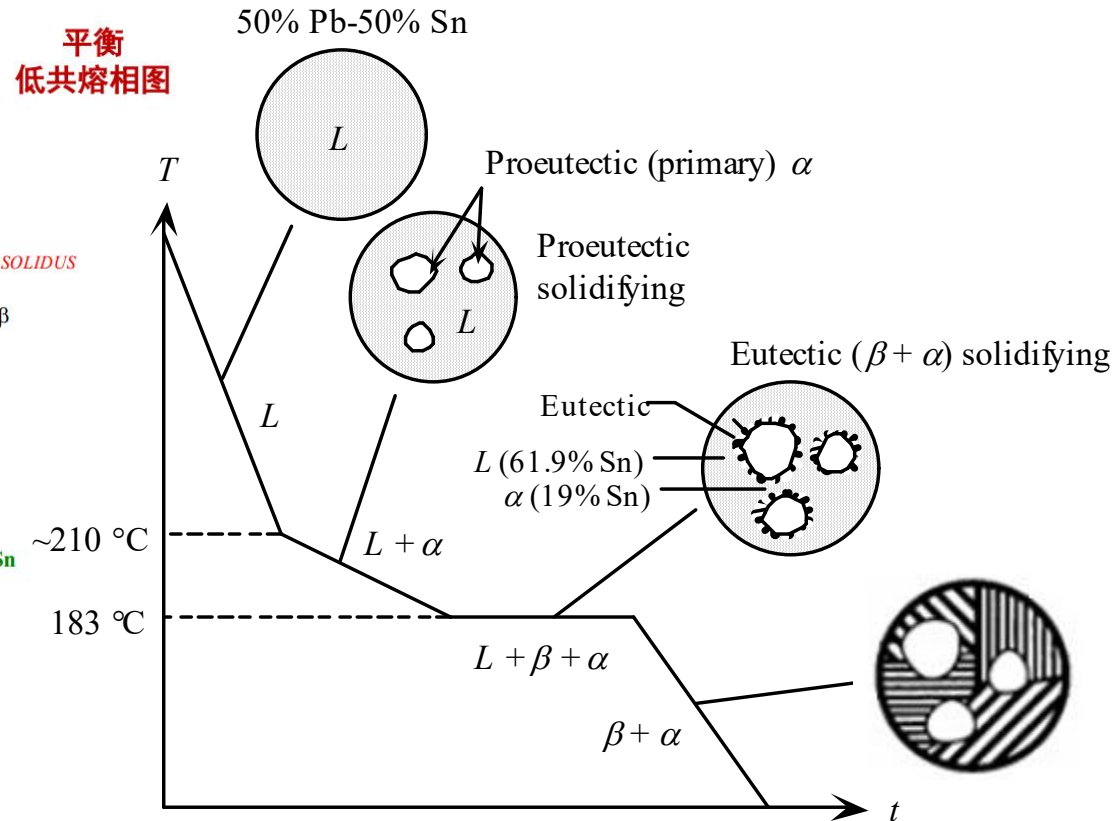
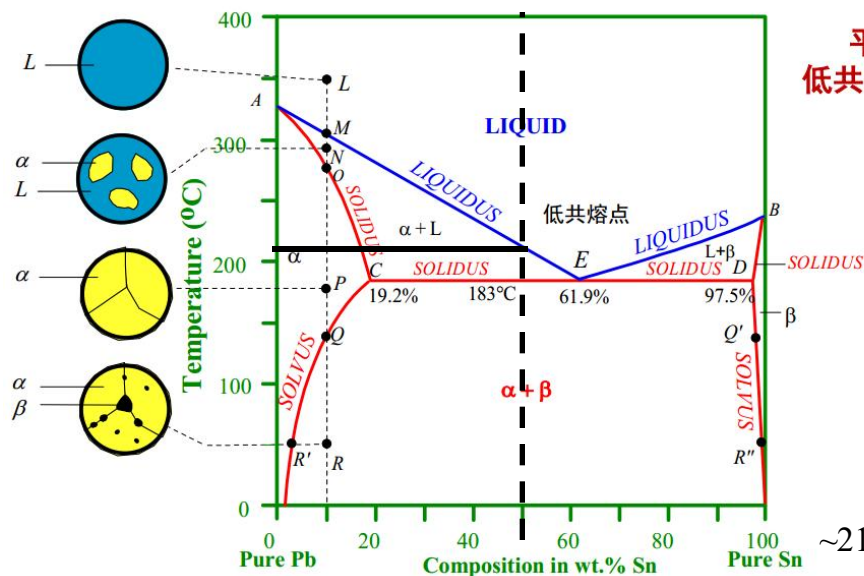
$$W'_{\beta} = \frac{C'_o - C_{\alpha}}{C'_{\beta} - C'_{\alpha}} = \frac{0.33 - 0.02}{1 - 0.02} = 0.316 \text{ } \beta \text{-phase}$$



1.32 Pb-Sn solder Consider the 50% Pb-50% Sn solder.

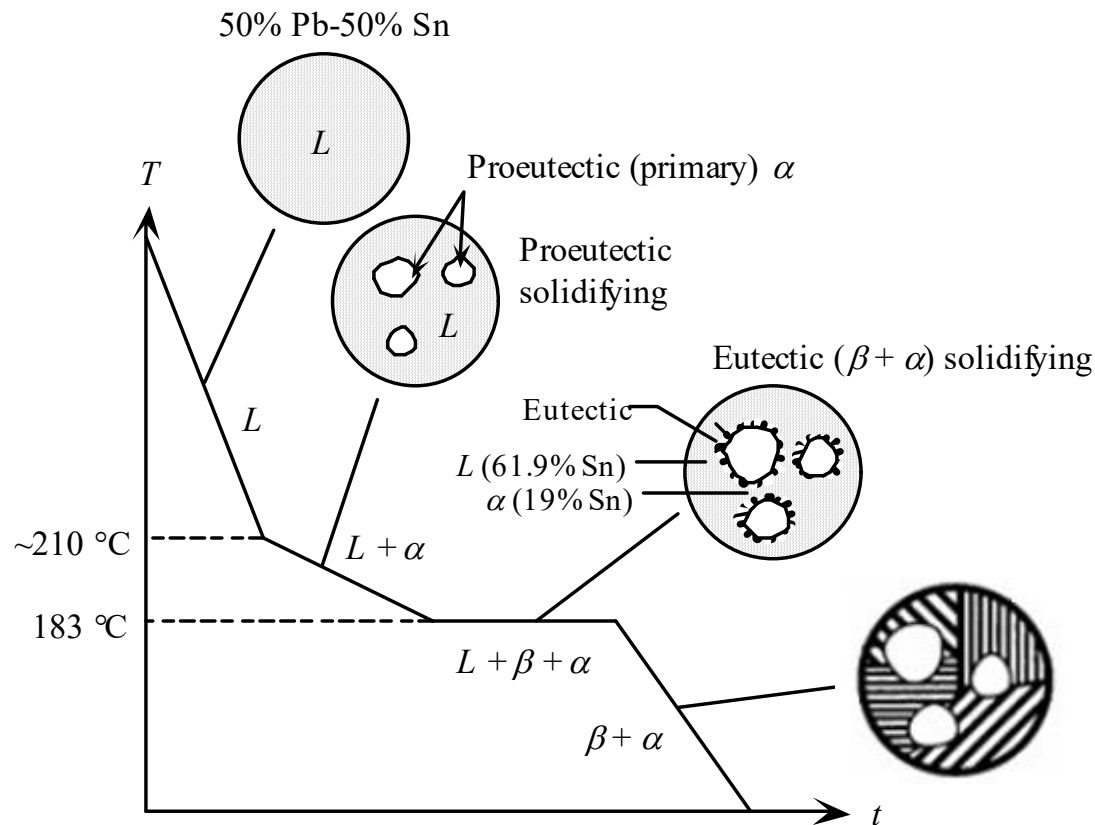
a. Sketch the temperature-time profile and the microstructure of the alloy at various stages as it is cooled from the melt.

Solution:



b. At what temperature does the solid melt?

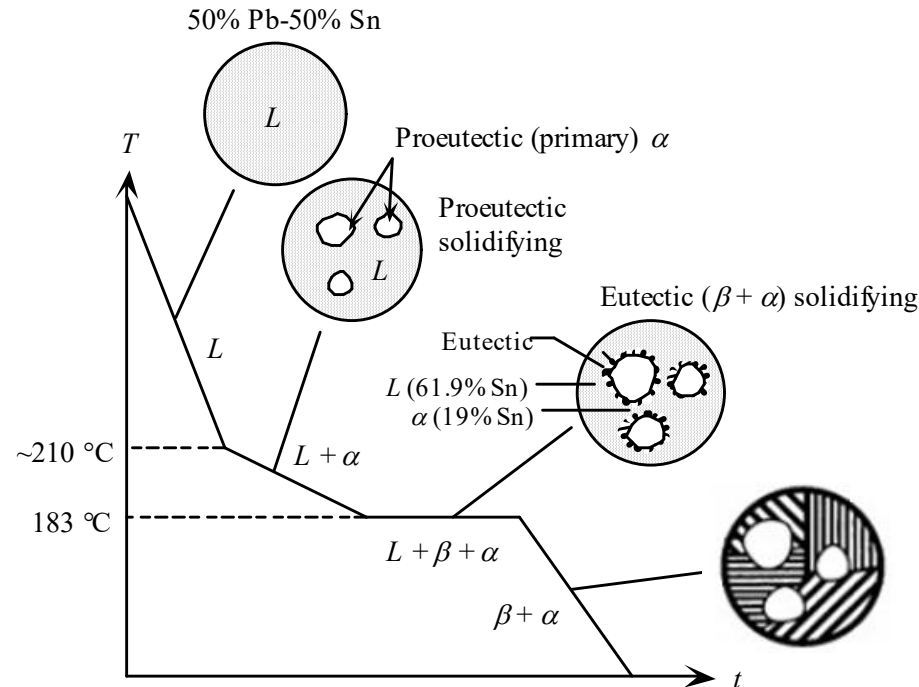
Solution:



When 50% Pb-50% Sn is cooled from the molten state down to room temperature, it begins to solidify at point A at about 210 °C. Therefore at 210 °C, the solid melt.

c. What is the temperature range over which the alloy is a mixture of melt and solid? What is the structure of the solid?

Solution:



Between 210°C and 183°C , the liquid has the eutectic composition and undergoes the eutectic transformation to become the eutectic solid. Below 183°C , the structure is a combination of the solid α -phase and the eutectic structure (which is composed of α and β layers).



2. α in the whole alloy.

Below 183 °C, the α and β phase coexist, and β is in the eutectic structure.

$$W_{\alpha 1} = \frac{C_{\beta} - C_o}{C_{\beta} - C_{\alpha 1}} = \frac{97.5 - 50}{97.5 - 19.2} = 0.607$$

Thus the mass of α in the whole solid is **0.607 kg**

3. α in the eutectic solid.

The mass of α in the whole solid is **0.607 kg**. Of this, 0.279 kg is in the primary (proeutectic) α phase. Thus $0.607 - 0.279 = \mathbf{0.328 \text{ kg}}$ of α is in the eutectic solid.

4. β in the alloy (Where is the β -phase?)

Because the solid is forming from a point where it consisted of the α and liquid phases only, the β -phase is present solely in the eutectic solid, with no primary β . Since the total mass of α in the solid is 0.607 kg, the remainder of the mass must be the β -phase. Thus, the mass of the β -phase is $1 \text{ kg} - 0.607 \text{ kg} = \mathbf{0.393 \text{ kg}}$.



e. Calculate **the specific heat** of the solder given the atomic masses of Pb (207.2) and Sn (118.71).

P7 Eq. 1.2

P25 e.g. 1.7

$$M_{AB} = n_A M_A + n_B M_B$$

Dulong-Petit rule: $c_s = \frac{3R}{M_{AB}}$

$$n_A = \frac{w_A/M_A}{w_A/M_A + w_B/M_B} \quad \text{and} \quad n_B = 1 - n_A$$

Solution:

$$n_{\text{Pb}} = \frac{\frac{w_{\text{Pb}}}{M_{\text{Pb}}}}{\frac{w_{\text{Pb}}}{M_{\text{Pb}}} + \frac{w_{\text{Sn}}}{M_{\text{Sn}}}} = \frac{\frac{0.5}{207.2}}{\frac{0.5}{207.2} + \frac{0.5}{118.71}} = 0.3642 \quad n_{\text{Sn}} = 1 - n_{\text{Pb}} = 1 - 0.3642 = 0.6358$$

$$c_s = \frac{3R}{M_{\text{PbSn}}} = \frac{3R}{n_{\text{Pb}} M_{\text{Pb}} + n_{\text{Sn}} M_{\text{Sn}}}$$

$$= \frac{3 \times (8.315 \text{ J K}^{-1} \text{ mol}^{-1})}{(0.3642) \times (207.2 \text{ g mol}^{-1}) + (0.6358) \times (118.71 \text{ g mol}^{-1})}$$

$$= 0.165 \text{ J K}^{-1} \text{ g}^{-1}$$



Homework 3



2.1 Electrical conduction

Na is a monovalent metal (BCC) with a density of 0.9712 g cm^{-3} . Its atomic mass is 22.99 g mol^{-1} . The drift mobility of electrons in Na is $53 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$.

a. Consider the collection of conduction electrons in the solid. If each Na atom donates one electron to the electron sea, *estimate the mean separation* between the electrons. (Note: if n is the concentration of particles, then the particles' mean separation $d = 1/n^{1/3}$.)

Solution:

the atomic concentration n_{at}

$$n_{at} = \frac{DN_A}{M_{at}} = \frac{(971.2 \text{ kg})(6.022 \times 10^{23} \text{ mol}^{-1})}{(22.99 \times 10^{-3} \text{ kg mol}^{-1})} = 2.544 \times 10^{28} \text{ m}^{-3}$$

$$d \approx \frac{1}{n_{at}^{1/3}} = \frac{1}{(2.544 \times 10^{28} \text{ m}^{-3})^{1/3}} = 3.40 \times 10^{-10} \text{ m or } 0.34 \text{ nm}$$



b. Estimate the mean separation between an electron (e^-) and a metal ion (Na^+), assuming that most of the time the electron prefers to be between two neighboring Na^+ ions. What is the approximate Coulombic interaction energy (in eV) between an electron and an Na^+ ion?

Solution

$$D = \frac{(\text{atoms in unit cell})(\text{mass of 1 atom})}{\text{volume of unit cell}} = \frac{2 \left(\frac{M_{\text{at}}}{N_A} \right)}{a^3}$$

$$a = \left[\frac{2M_{\text{at}}}{DN_A} \right]^{1/3} = \left[\frac{2(22.99 \times 10^{-3} \text{ kg mol}^{-1})}{(0.9712 \times 10^3 \text{ kg m}^{-3})(6.022 \times 10^{23} \text{ mol}^{-1})} \right]^{1/3}$$

$$= 4.284 \times 10^{-10} \text{ m or } 0.4284 \text{ nm}$$

$$R = (1/4) \sqrt{(3a^2)} = 1.855 \times 10^{-10} \text{ m or } 0.1855 \text{ nm}$$

The radius of the metal ion R and the lattice parameter a are related by $(4R)^2 = 3a^2$

the electrostatic potential energy

$$PE = \frac{(-e)(+e)}{4\pi\epsilon_0 d_{\text{electron-ion}}} = \frac{(-1.602 \times 10^{-19} \text{ C})(+1.602 \times 10^{-19} \text{ C})}{4\pi(8.854 \times 10^{-12} \text{ F m}^{-1})(1.855 \times 10^{-10} \text{ m})}$$

$$= -1.24 \times 10^{-18} \text{ J or } -7.76 \text{ eV}$$

P6



c. ① How does this electron/metal-ion interaction energy compare with the average thermal energy per particle, according to the kinetic molecular theory of matter? ② Do you expect the kinetic molecular theory to be applicable to the conduction electrons in Na? ③ If the mean electron/metal-ion interaction energy is of the same order of magnitude as the mean KE of the electrons, what is the mean speed of electrons in Na? ④ Why should the mean kinetic energy be comparable to the mean electron/metal-ion interaction energy?

Solution:

This electron-ion PE is **much larger than** the average thermal energy expected from the kinetic theory for a collection of “free” particles, that is $E_{\text{average}} = KE_{\text{average}} = 3(kT/2) \approx 0.039$ eV at 300 K. In the case of Na, **the electron-ion interaction is very strong** so we cannot assume that the electrons are moving around freely as if in the case of free gas particles in a cylinder. If we assume that the mean KE is roughly the same order of magnitude as the mean PE ,

$$KE_{\text{average}} = \frac{1}{2} m_e u^2 \approx |PE|_{\text{average}} = -1.24 \times 10^{-18} \text{ J}$$

$$\text{the mean speed } u \approx \left[\frac{2PE_{\text{average}}}{m_e} \right]^{1/2} = \left[\frac{2(1.24 \times 10^{-18} \text{ J})}{(9.109 \times 10^{-31} \text{ kg})} \right]^{1/2} \quad u = 1.65 \times 10^6 \text{ m/s}$$



There is a theorem in classical physics called the **Virial theorem** which states that if the interactions between particles in a system obey **the inverse square law** (as in Coulombic interactions) then the magnitude of the mean KE is **equal** to the magnitude of the mean PE. The Virial Theorem states that:

$$KE_{average} = -\frac{1}{2}PE_{average} \longrightarrow \text{Obviously, the mean speed } u \text{ is insensitive to the temperature.}$$

Therefore, the mean kinetic energy be comparable to the mean electron/metal-ion interaction energy.

d. Calculate the electrical conductivity of Na and compare this with the experimental value of $2.1 \times 10^7 \Omega^{-1} \text{ m}^{-1}$ and comment on the difference.

Solution:

If μ is the drift mobility of the conduction electrons and n is their concentration, then the electrical conductivity of Na is $\sigma = en\mu$. Assuming that each Na atom donates one conduction electron ($n = n_{at}$), we have

$$\sigma = en\mu = (1.602 \times 10^{-19} \text{ C})(2.544 \times 10^{28} \text{ m}^{-3})(53 \times 10^{-4} \text{ m}^2 \text{ V}^{-1} \text{ s}^{-1})$$

$$\sigma = 2.16 \times 10^7 \Omega^{-1} \text{ m}^{-1}$$



Homework 4



2.20 The Hall effect

Consider a rectangular sample, a metal or an n -type semiconductor, with a length L , width W , and thickness D . A current I is passed along L , perpendicular to the cross-sectional area WD . The face $W \times L$ is exposed to a magnetic field density B . A voltmeter is connected across the width, as shown in Figure 2.40, to read the Hall voltage V_H .

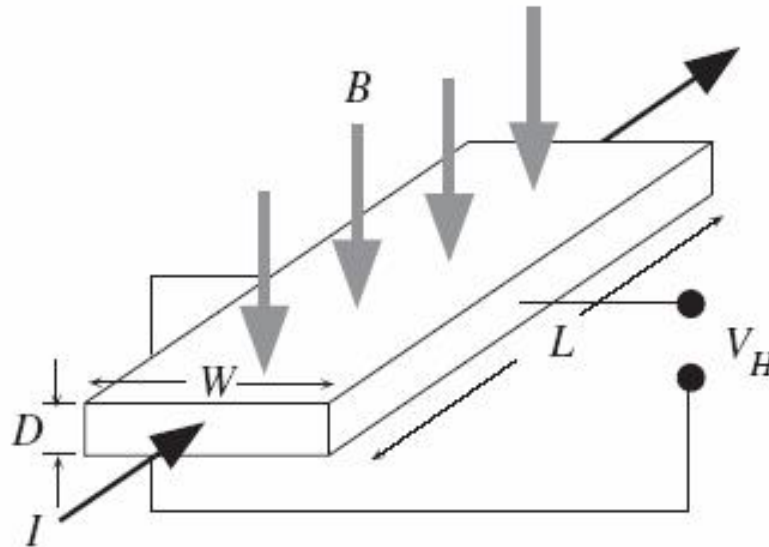


Figure 2.40 Hall effect in a rectangular material with length L , width W , and thickness D . The voltmeter is across the width W .

a. Show that the Hall voltage recorded by the voltmeter is

$$V_H = \frac{IB}{Den}$$

Solution:

The Hall coefficient, RH , is related to the electron concentration, n , by $RH = -1/(en)$, and is defined by $RH = E_y/(JB)$, where E_y is the electric field in the y -direction, J is the current density and B is the magnetic field. Equating these two equations:

$$-\frac{1}{en} = \frac{E_y}{JB} \quad \longrightarrow \quad E_y = -\frac{JB}{en}$$

This electric field is in the opposite direction of the Hall field (E_H) and therefore:

$$E_H = -E_y = \frac{JB}{en}$$

The current density perpendicular (going through) the plane $W \times D$ (width by depth)

$$J = \frac{I}{WD} \quad \longrightarrow \quad W = \frac{I}{JD}$$

The Hall voltage (V_H) across W is: $V_H = WE_H \quad \longrightarrow \quad V_H = \frac{IB}{Den}$



b. Consider a 1-micron-thick strip of **gold layer on an insulating substrate that is a candidate for a Hall probe sensor. If the current through the film is maintained at constant 100 mA, what is the magnetic field that can be recorded per μV of Hall voltage?**

Solution:

the gold's atomic mass: $M_{\text{at}} = 196.97 \text{ g mol}^{-1}$

the density: $d = 19.3 \text{ g cm}^{-3} = 19300 \text{ kg cm}^{-3}$

the concentration of free electrons:

$$n = \frac{dN_A}{M_{\text{at}}} = \frac{(19300 \text{ kg m}^{-3})(6.022 \times 10^{23} \text{ mol}^{-1})}{(196.97 \times 10^{-3} \text{ kg mol}^{-1})} = 5.901 \times 10^{28} \text{ m}^{-3}$$

the magnetic field B can be found using the equation for Hall voltage:

$$V_H = \frac{IB}{Den}$$

$$B = \frac{V_H Den}{I} = \frac{(1 \times 10^{-6} \text{ V})(1 \times 10^{-6} \text{ m})(1.602 \times 10^{-19} \text{ C})(5.901 \times 10^{28} \text{ m}^{-3})}{(0.1 \text{ A})}$$

$$B = 0.0945 \text{ T}$$



2.23 Temperature of a light bulb

P111 Eq. 2.10

a. Consider a 100 W, 120 V incandescent bulb (lamp). The tungsten filament has a length of 0.579 m and a diameter of 63.5 μm . Its resistivity at room temperature is 56 n Ω m. Given that the resistivity of the filament can be represented as

$$\rho = \rho_0 \left[\frac{T}{T_0} \right]^n$$

where T is the temperature in K, ρ_0 is the resistance of the filament at T_0 K, and $n = 1.2$, estimate the temperature of the bulb when it is operated at the rated voltage, that is, directly from the mains outlet. Note that the bulb dissipates 100 W at 120 V.



Solution:

First, find the current through the bulb at 100 W and 120 V.

$$R = \frac{U^2}{P} = \frac{(120 \text{ V})^2}{100 \text{ W}} = 144.0 \text{ } \Omega$$

The values for length of the filament ($L = 0.579 \text{ m}$) and diameter of the filament ($D = 63.5 \text{ } \mu\text{m}$) at operating temperature are given. Using these values we can find the resistivity of the filament when the bulb is on (ρ_1).

$$R = \frac{\rho_{eff} L}{A} = \frac{\rho_1 L}{\frac{\pi}{4} D^2}$$

$$\Rightarrow \rho_1 = \frac{R \frac{\pi}{4} D^2}{L} = \frac{(144.0 \text{ } \Omega) \times \frac{\pi}{4} \times (63.5 \times 10^{-6} \text{ m})^2}{(0.579 \text{ m})} = 7.876 \times 10^{-7} \text{ } \Omega\text{m}$$



Now the bulb's operating temperature (T_1) can be found using our obtained values in the equation for resistivity of W (assuming room temperature $T_o = 293$ K and given $n = 1.2$):

$$\rho = \rho_0 \left[\frac{T}{T_0} \right]^n \quad \text{P108 Eq. 2.20}$$

$$T_1 = T_0 \left[\frac{\rho_1}{\rho_0} \right]^{\frac{1}{n}} = (293 \text{ K}) \times \left[\frac{7.876 \times 10^{-7} \text{ } \Omega \text{ m}}{56 \times 10^{-9} \text{ } \Omega \text{ m}} \right]^{\frac{1}{1.2}} = 2652 \text{ K}$$



b. Suppose that the electrical power dissipated in the tungsten wire is totally radiated from the surface of the filament. The radiated power at the absolute temperature T can be described by Stefan's Law

$$P_{\text{radiated}} = \varepsilon \sigma_s A (T^4 - T_o^4)$$

where σ_s is Stefan's constant ($5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$), ε is the emissivity of the surface (0.35 for tungsten), A is the surface area of the tungsten filament, and T_o is the room temperature (293 K). Obviously, for $T > T_o$, $P_{\text{radiated}} = \varepsilon \sigma_s A T^4$.

Assuming that all of the electrical power is radiated from the surface, estimate the temperature of the filament and compare it with your answer in part (a).



Solution:

First we need the surface area A of the Tungsten filament. Since it is cylindrical in shape:

$$A = L(\pi D) = (0.579 \text{ m}) \times \pi \times (63.5 \times 10^{-6} \text{ m}) = 0.0001155 \text{ m}^2$$

Now the temperature of the filament T_1 can be found by isolating it in Stefan's equation and substituting in the given values for emissivity ($\varepsilon = 0.35$), Stefan's constant ($\sigma_s = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$) and room temperature ($T_o = 293 \text{ K}$).

$$T_1 = \left(\frac{P}{\varepsilon \sigma_s A} + T_o^4 \right)^{\frac{1}{4}} = \left[\frac{100 \text{ W}}{0.35 \times (5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-1}) \times 0.0001155 \text{ m}^2} + (293 \text{ K})^4 \right]^{\frac{1}{4}}$$
$$= 2570 \text{ K}$$

Note: We can even ignore T_o to get the same temperature since $T_o \ll T_1$:

$$P = \varepsilon \sigma_s A T_1^4 \quad T_1 = \left(\frac{P}{\varepsilon \sigma_s A} \right)^{\frac{1}{4}} = \left[\frac{100 \text{ W}}{(0.35)(5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-1})(0.0001155 \text{ m}^2)} \right]^{\frac{1}{4}}$$

$$T_1 = 2570 \text{ K}$$

These values are fairly close to the answer obtained in part (a).



- c. If the melting temperature of W is 3407 °C, what is the voltage that guarantees that the light bulb will blow?

Solution:

Let V be the voltage and R be the resistance when the filament is at temperature T_m . We are given the temperature $T_m = 3407\text{ °C} + 273 = 3680\text{ K}$. Since we know the following:

$$R = \frac{L}{\frac{\pi}{4} D^2} \rho \qquad \rho = \rho_0 \left[\frac{T_m}{T_0} \right]^n$$

We can make a substitution for ρ and use the values given for the light bulb filament to find the resistance of the filament at temperature T_m .

$$\begin{aligned} R &= \frac{L}{\frac{\pi}{4} D^2} \rho_0 \left[\frac{T_m}{T_0} \right]^n = \frac{0.579\text{ m}}{\frac{\pi}{4} \times (63.5 \times 10^{-6}\text{ m})^2} \times (56 \times 10^{-9}\text{ }\Omega\text{ m}) \times \left[\frac{3680\text{ K}}{293\text{ K}} \right]^{1.2} \\ &= 213.3\text{ }\Omega \end{aligned}$$



Assuming that all electrical power is radiated from the surface of the bulb, we can use Stefan's law and substitute in V^2/R for power P :

$$\frac{U^2}{R} = \varepsilon \sigma_s A (T_1^4 - T_0^4)$$

$$U^2 = R \varepsilon \sigma_s A (T_m^4 - T_0^4)$$

$$U^2 = 213.3 \, \Omega \times 0.35 \times 5.67 \times 10^{-8} \, \text{W m}^{-2} \text{K}^{-4} \times 0.0001155 \, \text{m} \times \left[(3680 \, \text{K})^4 - (293 \, \text{K})^4 \right]$$

$$\Rightarrow U = 299 \, \text{V}$$



2.24 Einstein relation and ionic conductivity

In the case of ionic conduction, ions have to jump from one interstice to the neighboring one. This process involves overcoming a potential energy barrier just like atomic diffusion, and drift and diffusion are related. The drift mobility μ of ions is proportional to the diffusion coefficient D because drift is limited by the atomic diffusion process. The Einstein relation relates the two by

$$\frac{D}{\mu} = \frac{kT}{e}$$

The diffusion coefficient of the Na^+ ion in sodium silicate ($\text{Na}_2\text{O-SiO}_2$) glasses at 400°C is $3.4 \times 10^{-9} \text{ cm}^2 \text{ s}^{-1}$. The density of such glasses is approximately 2.4 g cm^{-3} . Calculate **the ionic conductivity and resistivity** of (17.5 mol% Na_2O)(82.5 mol% SiO_2) sodium silicate glass at 400°C and compare your result with the experimental value of about $10^4 \Omega \text{ cm}$ for the resistivity.

Solution:

The atomic mass of one unit is

$$M_{at} = 0.175[2 \times (23) + 16] + 0.825[28.1 + 2 \times (16)] = 60.43 \text{ g mol}^{-1} \text{ of } [(\text{Na}_2\text{O})_{0.175}(\text{SiO}_2)_{0.825}]$$



The number of $[(\text{Na}_2\text{O})_{0.175}(\text{SiO}_2)_{0.825}]$ units per unit volume

$$n = \frac{d N_A}{M_{at}} = \frac{(2.4 \text{ g cm}^{-3})(6.02 \times 10^{23} \text{ mol}^{-1})}{(60.43 \text{ g mol}^{-1})} = 2.39 \times 10^{22} \text{ units cm}^{-3}$$

Each of these $[(\text{Na}_2\text{O})_{0.175}(\text{SiO}_2)_{0.825}]$ units has 0.175×2 number of Na atoms so that Na^+ -ion concentration is

$$n_i = 0.175 \times 2 \times 2.39 \times 10^{22} \text{ cm}^{-3} = 8.37 \times 10^{21} \text{ cm}^{-3}$$

The mobility of sodium ions

$$\mu_i = \frac{eD}{kT} = \frac{(1.602 \times 10^{-19} \text{ C})(3.4 \times 10^{-9} \text{ cm}^2 \text{ s}^{-1})}{(1.38 \times 10^{-23} \text{ J K}^{-1})(400 + 273) \text{ K}} = 5.89 \times 10^{-8} \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$$

Thus the conductivity of the glass

$$\sigma = en_i \mu_i = (1.602 \times 10^{-19} \text{ C})(8.37 \times 10^{21} \text{ cm}^{-3})(5.89 \times 10^{-8} \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}) = 7.90 \times 10^{-5} \Omega^{-1} \text{ cm}^{-1}$$

P135 Eq. 2.51

The resistivity $\rho = \frac{1}{\sigma} = \frac{1}{7.90 \times 10^{-5} \Omega^{-1} \text{ cm}^{-1}} = 1.27 \times 10^4 \Omega \text{ cm}$



THANKS !

